Final Project – ML course 2026/27

Dataset description

The provided dataset consists of data from third year undergraduate students at the Medical School at the University of Navarra. Several features, including basic data such as age and sex, as well as metrics known to affect the development of Myopia are provided for each student.

With the provided data the goal is to make four types of predictions:

- **Binary classification** to predict whether a student has myopia (M) or not. This is encoded in column M, with labels YES and NO.
- **Binary classification** to predict whether a student has myopia magna (MM) or not. This is encoded in column MM, with labels YES and NO. (MM es más de 6 dioptrias)
- **Multi-Label classification**, to combine both previous columns into one, denoted as 'COMBO', where a student is classified as M (myopia), MM (myopia magna), or C (control or no myopia).
- **Multi-Label classification**, specified in column 'DCOMBO', where myopia students are further classified as type M1 or M2.

A training dataset is provided to train the models, as well as a testing dataset WITHOUT LABELS. The ultimate goal is to produce labels in the test dataset that are as accurate as possible, but all the process to achieve that goal will count towards the grade. You can also use libraries/functions/architectures not seen in class, although you should understand them. This will actually count positively! Just imagine you are in a company and are given this dataset, and at the end of the project you have to present your work: will you just show the accuracy on the validation set?

Group: In pairs (one group of 1 or 3 people if there is an odd number)

Deliverables: A zip folder containing at least the following files:

1. The UNIQUE inferred labels for the test dataset stored in column**s** in a .TXT format. Even if more than one model is presented, the final test labels should correspond only to the model that you believe is performing the best **(you can use different models for the different classification tasks).**

2. A Jupyter notebook (or Python scripts) serving as a report, i.e., the notebook should be commented (in English!). The notebooks should run with no errors. Nevertheless, you should submit it already executed so that the output cells are visible.

3. All files needed for the program to run correctly without errors (except for the training and test data which there are already available).

4. The slides that you will use for the presentation. The slides should be prepared as if the work was going to be presented in a conference (company), where you briefly introduce the problem, how you tackled it, the methodology employed, the results you obtained, and the conclusions.

5. At the end of the presentation you should include some slides where you specify: the work led by each of the members of the team (or which parts were done together); if the work load split was 50/50 or different; if you used Kears/PyTorch or both; and how you used generative AI models throughout the work (e.g., to generate code, to get ideas about how to proceed, etc.).

6. A **10' video** with the recorded presentation.

Submission: Via ADI

Deadline: Friday November 6 before 11.59pm.

Evaluation: The following aspects will be evaluated in the project:

1. Presentation.

2. Accuracy achieved in the test.

3. Methodology employed.

4. A well-written report (within the Jupyter notebook) that justifies all actions and functions.